Understanding Options

and the Options Payoff Visualizer

A beginner's guide - no finance knowledge needed

IS4350 Finance Engineering | University of Moratuwa

Part	What you will learn
1	What an option is (everyday example)
2	Call options and put options
3	Key words explained
4	Payoff vs profit/loss, with a full numeric example
5	Why options exist and who uses them
6	What our system does, how to use it, and how to read the graphs
7	Self-test and a one-minute demo introduction

1. What is an option?

Imagine you want to buy a concert ticket that costs **$100** today. You think the band is getting famous, so the price may rise next month. You pay the seller a small fee of **$5** to *lock in* the right to buy that ticket at **$100** next month, whatever happens to the market price.

Next month the ticket costs...	What you do	Result
$120	Use your right and buy at $100	You save $20, minus the $5 fee = **+$15**
$90	Do not use it; buy at $90 on the market	You lose only the **$5 fee**

> *An option is a contract. You pay a small fee now, and in return you get the RIGHT (not the duty) to buy or sell something at a fixed price later. As the buyer, the most you can lose is the fee.*

2. Call options and put options

Call option = the right to BUY

You use a call when you think the price will **go up**. Example: a phone costs $100. You pay $5 for the right to buy it at $100 in a month.

Price in a month	What you do	Result
$130	Use your right (buy at $100)	+$30 - $5 fee = **+$25**
$105	Use your right	$5 - $5 fee = **0** (break-even)
$90	Do not use it	**-$5** (only the fee)

Put option = the right to SELL

You use a put when you think the price will **go down**, or when you want protection (like insurance). Example: you own a phone worth $100 and fear its price will drop. You pay $4 for the right to sell it at $100 in a month.

Price in a month	What you do	Result
$70	Use your right (sell at $100)	+$30 - $4 fee = **+$26**
$96	Use your right	$4 - $4 fee = **0** (break-even)
$110	Do not use it	**-$4** (only the fee)

Side by side

	Call	Put
Right to	**Buy**	**Sell**
Buyer hopes price goes	Up	Down
Buyer's biggest loss	The premium	The premium
Buyer's biggest gain	Unlimited	Strike - premium
Seller's biggest gain	The premium	The premium
Seller's biggest loss	Unlimited	Strike - premium
Break-even price	Strike + premium	Strike - premium
Everyday comparison	Reserving at today's price	Insurance

Memory trick: with a **call** you 'call' the asset to you (buy) and want the price up. With a **put** you 'put' the asset onto someone else (sell) and want the price down.

Every option has two sides. Calls and puts both have a buyer and a seller. A put is not 'the seller's option': people buy puts all the time, for example to protect shares they already own.

3. Key words

Word	Simple meaning	Example
Strike price (K)	The locked-in price in the contract	$100
Premium (P)	The fee the buyer pays for the right	$5
Market price at maturity (S_T)	The real price on the last day	$120 or $90
Maturity / expiry	The date the contract ends	In one month
Buyer (long)	Pays the fee, holds the right. Loss is limited to the fee.	You
Seller (short / writer)	Takes the fee, must honour the deal if the buyer uses it. Gain is limited to the fee; loss can be large.	The other side

The **seller is the exact opposite of the buyer**: if the buyer gains $15, the seller loses $15. If the buyer loses the $5 fee, the seller keeps that $5.

4. Payoff vs profit/loss

- **Payoff** = what the option is worth on the last day, *ignoring* the fee. It is never negative for the buyer.
- **Profit/Loss** = payoff minus the premium paid. It can be negative.

Formulas (per 1 unit, at maturity)

	Call	Put
Payoff (buyer)	$\max(S_T - K, 0)$	$\max(K - S_T, 0)$
Profit (buyer)	Payoff - P	Payoff - P
Profit (seller)	-(Payoff - P)	-(Payoff - P)
Break-even	K + P	K - P
Buyer max gain / seller max loss	Unlimited	K - P
Buyer max loss / seller max gain	P	P

max(a, 0) simply means 'a, but never below zero'. If the option is not worth using, the payoff is zero.

Full example: call option, K = 100, P = 5

S_T	Payoff	Buyer profit	Seller profit
120	20	+15	-15
105	5	0 (break-even)	0
102	2	-3	+3
90	0	-5 (max loss)	+5 (max gain)

Full example: put option, K = 100, P = 4

S_T	Payoff	Buyer profit	Seller profit
80	20	+16	-16
96	4	0 (break-even)	0
98	2	-2	+2
110	0	-4 (max loss)	+4 (max gain)

Look at the 102 and 98 rows: the option pays something, but the buyer still loses because the payoff is smaller than the premium paid.

5. Why do options exist?

- **Insurance (hedging):** an airline can lock in a fuel price so a price jump does not ruin its year. A farmer can lock in a minimum selling price for a crop.
- **Investing and speculating:** a trader can bet on a price rising or falling while risking only the small fee.
- **Managing risk:** banks and companies use options to limit how much they can lose.
- **Learning:** options are a standard topic in finance courses because they show how risk and reward work.

6. What is our system?

The Options Payoff Visualizer is a small web app. You type in four numbers, and it instantly shows how much the buyer and the seller of a call option and a put option would gain or lose, draws a graph for each, and lists the results at a range of market prices.

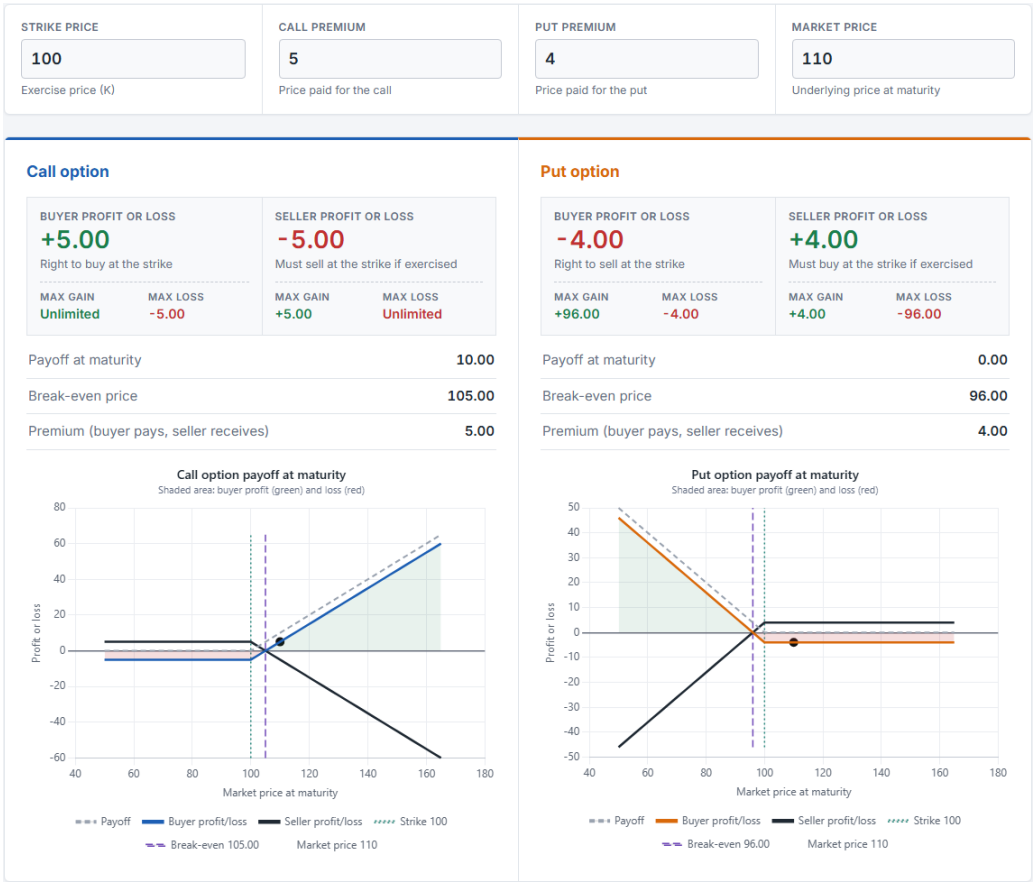
Why we built it

Our assignment (Group 3: Options) asks us to build a tool that calculates profits and losses for call and put options and displays payoff graphs. Building it forces us to understand the formulas, and the graphs make a confusing topic easy to see.

The four inputs

Input	Meaning	Default value
Strike price	The locked-in price (K), shared by both options	100
Call premium	The fee paid for the call	5
Put premium	The fee paid for the put	4
Market price	The real price on the last day (S_T), shared by both options	110

The strike and the market price are shared, so the call and the put can be compared on the same asset side by side.



The input row and the call and put sections with the default values.

What it shows you

- **Buyer profit/loss** and **seller profit/loss** next to each other, at the same size, with **max gain** and **max loss** for each side.
- **Payoff** at maturity, **break-even price** and the **premium** (paid by the buyer, received by the seller).
- A **graph** for the call and a graph for the put.
- A **scenario table** with the profit or loss of all four positions (call buyer, call seller, put buyer, put seller) at prices from 70% to 130% of the strike.

Gains are shown in green with a + sign and losses in red with a - sign, so the result is clear even without colour.

The scenario table

The row at the strike is tagged STRIKE and the row at the current market price is highlighted and tagged CURRENT. **Click any row** (or select it and press Enter) to use that price; every number and both graphs update. If you type a market price that is not in the list, it is added as its own row.

Profit or loss at different market prices — Click a row to use that price.

MARKET PRICE	CALL BUYER	CALL SELLER	PUT BUYER	PUT SELLER
70.00	-5.00	+5.00	+26.00	-26.00
80.00	-5.00	+5.00	+16.00	-16.00
90.00	-5.00	+5.00	+6.00	-6.00
95.00	-5.00	+5.00	+1.00	-1.00
100.00 STRIKE	-5.00	+5.00	-4.00	+4.00
105.00	0.00	0.00	-4.00	+4.00
110.00 CURRENT	+5.00	-5.00	-4.00	+4.00
120.00	+15.00	-15.00	-4.00	+4.00
130.00	+25.00	-25.00	-4.00	+4.00

Scenario table with K = 100, call premium 5, put premium 4 and market price 110.

Input checks

The strike must be greater than 0, and the premiums and the market price must be 0 or more. A blank or invalid value shows a red message instead of the results. Nothing crashes, and the results come back as soon as the value is fixed.

How it works (workflow)

You type K, the premiums and S_T → the calculation module (`options.js`) applies the formulas → the screen shows the numbers → the chart module (`PayoffChart.jsx`) draws the graphs → you change a number or click a scenario row and everything updates instantly.

How to read the graphs

Line / mark	Meaning
Grey dashed line	Payoff (before subtracting the fee)
Blue line (call graph) / orange line (put graph)	Buyer's profit or loss
Dark solid line	Seller's profit or loss (the upside-down mirror of the buyer's line)
Green shaded area	Prices where the buyer makes a profit
Red shaded area	Prices where the buyer makes a loss
Teal dotted vertical line	Strike price
Purple dashed vertical line	Break-even price
Black dot	The market price you entered, on the buyer's line

Hover over a graph to see the payoff, buyer and seller values at that price. Click a name in the legend to hide or show that line.

Call graph: the buyer's line is flat at minus-premium on the left (you lose only the fee), crosses zero at the break-even price, then rises steadily - a 'hockey stick'. **Put graph:** the same idea flipped: profit grows as the price falls, and the loss is flat at minus-premium on the right.

Try it yourself

Test	Inputs	Expected result
Call, price above strike	K 100, call premium 5, market 120	Payoff 20, buyer +15, seller -15, break-even 105
Put, price below strike	K 100, put premium 4, market 80	Payoff 20, buyer +16, seller -16, break-even 96
Call, price below strike	K 100, call premium 5, market 90	Payoff 0, buyer -5, seller +5
Scenario row	Click the 80.00 row	Market price becomes 80, put buyer +16
Bad input	Blank field or strike = 0	Red error message, no crash

7. Self-test and demo introduction

Self-test (call option, K = 100, premium = 5)

- The price ends at 130. Buyer's profit? **25** (payoff 30 minus fee 5).
- The price ends at 80. Buyer's profit? **-5** (only the fee is lost).
- At what price do you break even? **105** (strike plus fee).
- If the buyer makes 25, what does the seller make? **-25**.
- What is the most the call seller can lose? **No limit**, because the price can keep rising.

One-minute introduction for the demo

An option is a contract that gives you the right, but not the obligation, to buy or sell something at a fixed price on a future date, in return for a small fee called the premium. A call is the right to buy; a put is the right to sell. Our application takes the strike price, the call and put premiums and the market price at maturity. It calculates the payoff and the profit or loss for the buyer and the seller of each option, shows their best and worst cases, draws the payoff graphs and lists the results across a range of prices. Let us show you how it works.

Remember

- The buyer's loss is limited to the **premium**.
- The seller's result is always the **mirror image** of the buyer's.
- Payoff ignores the fee; **profit/loss includes it**.
- Call break-even = **strike + premium**; put break-even = **strike - premium**.